Blind Faith in Text: Evaluating Multi-modal Embedding Robustness in Remote Sensing Land Cover Analysis

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I. Literature Survey

The transformer architecture [1] and its vision adaptation ViT [2] established the foundation for modern Vision-Language Models (VLMs). After the adoption of transformer architecture for both models, Li et al. [3] investigates whether we can adopt the multi-modal embeddings on a down-stream task like remote sensing and shows that the multi-modal models can achieve good results, even better than the CNNs, which is highly adopted on the remote sensing domain. There are several attempts to adopt the technology for remote sensing, one example is GeoRSCLIP [4], demonstrating strong zero-shot performance on RS tasks. At a larger scale, EarthGPT [5] and GeoChat [6] extend VLMs to instruction-following for RS applications. There are lots of research for adapting the VLMs on remote-sensing domain, but before that, The gap VLMs possess is not studied enough. Deng et al. [7] identify a "blind faith in text" phenomenon in general VLMs, where models trust textual input over visual evidence under modality conflict, leading to significant performance drops. And VLM models use different fusion techniques to fuse both modalities on a common space, and it leads to 2 different architectures: dual-encoder VLMs and fusion-encoder VLMs [3]. Dual-encoder uses 2 different encoders to generate embeddings for both modalities, and measure similarity with a dot product, while fusion-encoders uses cross-attention to allow deeper relationships. These different architectural choices and their effects on the "blind faith in text" phenomenon is an unexplored study.

II. Research Question and Proposal

This project investigates whether the blind faith in text phenomenon happens in RS land cover analysis using multi-modal embeddings. The research question is *Does caption accuracy affect image-text alignment in RS multi-modal embeddings, and does fusion strategy moderate this effect?*

Using the provided dataset of 10,000 RS images with seven land cover classes (tree, shrub, grass, crop, built-up, barren, water) and associated captions, we propose evaluating RemoteCLIP [8] as a dual-encoder baseline under three conditions: (1) image-only, (2) image with correct caption, and (3) image with misleading caption. A cross-attention fusion module with frozen RemoteCLIP encoders will be compared against the dual-encoder baseline to determine whether fusion strategy increases or reduces text bias. Performance is measured via cosine similarity scores and classification accuracy across land cover classes.

III. Proof of Concept

Feasibility was demonstrated using CLIP [9] on image 0073.png from the provided dataset, which contains a 92% grassland scene. Two caption conditions were evaluated: a correct caption describing grassland and a misleading caption taken from another sample's caption describing urban infrastructure. As shown in Figure 1, correct caption produced the highest cosine similarity (0.2672), followed by misleading (0.2571). This confirms that caption quality measurably affects embedding alignment but not significantly. Phase 2 will replace CLIP with RemoteCLIP for RS-specific evaluation on the full dataset and will be a broader approach.

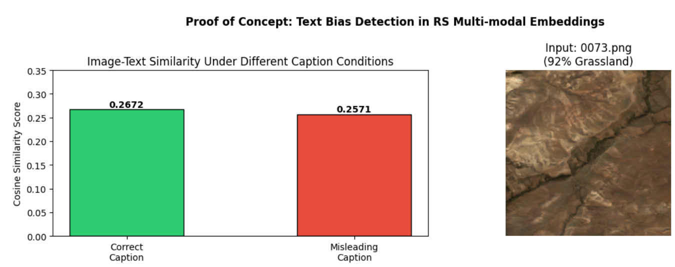

Fig. 1. Cosine similarity scores under correct and misleading caption conditions for a grassland RS image (Phase 1 PoC).

IV. Methods

The dataset used in this study is a subset of ARAS400k [10] with 10,000 samples, a remote sensing segmentation and captioning dataset accepted at CVPR 2026, containing 100,240 real and 300,000 synthetic RS images with land-cover segmentation maps and over 2 million captions generated by VLMs.

Two pre-trained models are evaluated: CLIP [9] as the general-purpose baseline and RemoteCLIP [8] as the RS-specific model. Both follow a dual-encoder architecture, encoding images and text independently before computing cosine similarity in a shared embedding space.

Three experiments are done on the full 10,000 image dataset. First, *Image-only Baseline*: each image embedding is compared against seven land cover class description templates, and the maximum similarity is recorded as the image-only score. Second, *Caption Similarity*: cosine similarity is computed between each image and two caption types (correct (the image's own caption from the hybrid_gemma3-4b column) and misleading (a caption randomly sampled from images whose dominant land cover class differs from the query image, seed=42)). Third, *Zero-shot Classification*: each image is classified into one of seven land cover classes by comparing its embedding against class description templates and selecting the highest similarity match. Dominant land cover class is determined from the composition percentage columns.

V. Results and Benchmarking

Table I reports average cosine similarity scores across all 10,000 images. Both models show highest similarity in the image-only condition, where no textual input is introduced. Notably, even correct captions reduce similarity compared to the image-only baseline, suggesting that textual input lowers the accuracy regardless of caption accuracy. This is consistent with the blind faith in text phenomenon identified by Deng et al. [7], confirming that the effect is also seen in RS multi-modal embeddings.

RemoteCLIP shows a larger drop from image-only to misleading captions (0.0306) compared to CLIP (0.0141), suggesting that RS-specific fine-tuning increase text bias. This is an important finding for RS applications where metadata quality cannot be guaranteed.

TABLE I

Average cosine similarity scores across 10,000 RS images under three conditions. Image-only serves as the baseline.

Model	Image-only	Correct	Misleading
CLIP	0.2804	0.2788	0.2663
RemoteCLIP	0.2494	0.2430	0.2188

Zero-shot land cover classification results are shown in Figure 2. CLIP achieves 17.13% accuracy while RemoteCLIP achieves 15.95%, both above random chance (14.28% for 7 classes). The lower accuracy of RemoteCLIP suggests that RS-specific fine-tuning optimizes for retrieval alignment rather than zero-shot classification, consistent with catastrophic forgetting phenomenon observed in fine-tuned models.

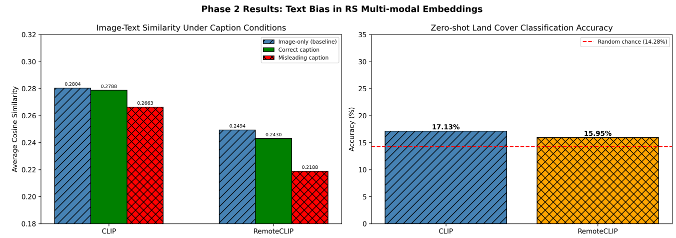

Fig. 2. Left: Average cosine similarity under image-only, correct, and misleading caption conditions for CLIP and RemoteCLIP. Right: Zero-shot land cover classification accuracy. RemoteCLIP accuracy decreases more after caption corruption despite being RS-specific.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[2] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations*, 2021.

[3] X. Li, C. Wen, Y. Hu, Z. Yuan, and X. X. Zhu, "Vision-language models in remote sensing: Current progress and future trends," *IEEE Geoscience and Remote Sensing Magazine*, vol. 12, no. 2, pp. 32–66, 2024.

[4] Z. Zhang, T. Zhao, Y. Guo, and J. Yin, "Rs5m and georsclip: A large-scale vision-language dataset and a large vision-language model for remote sensing," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–23, 2024.

[5] W. Zhang, M. Cai, T. Zhang, Y. Zhuang, and X. Mao, "Earthgpt: A universal multimodal large language model for multisensor image comprehension in remote sensing domain," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–20, 2024.

[6] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "Geochat: Grounded large vision-language model for remote sensing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 831–27 840.

[7] A. Deng, T. Cao, Z. Chen, and B. Hooi, "Words or vision: Do vision-language models have blind faith in text?" in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025, pp. 3867–3876.

[8] F. Liu, D. Chen, Z. Guan, X. Zhou, J. Zhu, Q. Ye, L. Fu, and J. Zhou, "Remoteclip: A vision language foundation model for remote sensing," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1–16, 2024.

[9] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning*, 2021, pp. 8748–8763.

[10] M. Çağlar and A. Temizel, "Aras400k: A remote sensing segmentation and image caption dataset," *arXiv preprint arXiv:2603.09625*, 2026.